AIML: Adaptive Importance Metric Learning for Reinforcement Learning

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Abstract

We study the problem of learning how the importance of each observation dimension changes depending on the trajectory in reinforcement learning (RL). Existing RL methods use either fixed or implicitly learned weights over observation dimensions, making it difficult to interpret their importance across trajectories, particularly in partially observable and stochastic environments. We propose Adaptive Importance Metric Learning (AIML), a framework that learns trajectory-dependent importance weights over observation dimensions. The framework aggregates trajectory information to produce dimension-wise importance weights that explicitly scale each observation dimension, transforming the current observation into a context-dependent representation. The proposed approach is simple and modular, allowing it to be easily integrated into standard actor-critic methods with minimal architectural changes. Empirically, the proposed approach improves sample efficiency and final performance when the underlying architecture does not sufficiently preserve trajectory-dependent, dimension-wise relevance. Furthermore, our analysis suggests that it can provide an additional interpretability lens relative to standard representation learning approaches.

1 Introduction

Reinforcement learning (RL) has achieved remarkable success in a wide range of domains, largely driven by advances in representation learning [Mnih et al., 2015] for high-dimensional observations. Modern RL agents typically rely on neural networks to encode observations into latent state representations[Schulman et al., 2017], which are then used for policy and value estimation. Despite these advances, most existing approaches assume that the importance of observation dimensions is either fixed or implicitly determined by the model parameters.

However, in many real-world environments, the relevance of observation components is not fixed but varies across trajectories. This challenge arises already at the stage of observation design. In reinforcement learning, observations are typically normalized using predefined bounds, and subsets of observation dimensions are grouped or selected based on prior knowledge or extensive empirical tuning. Furthermore, determining the relative importance of different observation components commonly relies on heuristics, insights from underlying physical models, or large-scale experimental search. When such scaling or weighting is mis-specified, its impact can be amplified in partially observable settings [Igl et al., 2018], where observations are inherently incomplete, and small distortions in importance can affect the interpretation of the underlying state. Moreover, when importance is handled only implicitly through representation learning, the lack of explicit parameterization makes it difficult to interpret how the contribution of individual observation dimensions varies across trajectories.

Existing reinforcement learning approaches such as deep q-network (DQN) [Mnih et al., 2015] and proximal policy optimization (PPO) [Schulman et al., 2017] rely on neural representations of

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observations for value estimation or policy optimization, where observations are directly mapped to latent features through feedforward networks. To address partial observability, recurrent reinforcement learning methods such as deep recurrent q-networks (DRQN) [Hausknecht and Stone, 2015] and recurrent replay distributed dqn (R2D2) [Kapturowski et al., 2019] utilize lstm-based [Hochreiter and Schmidhuber, 1997] architectures to process observation histories, summarizing past observations into hidden states that serve as implicit belief representations. More recently, transformer-based approaches, including decision transformer [Chen et al., 2021], model observation-action trajectories as sequences and capture long-range dependencies through self-attention mechanisms, enabling more expressive temporal reasoning over observations. Beyond transformers, newer sequence models such as state space models (e.g., mamba [Gu and Dao, 2023], decision mamba [Lv et al., 2024]) have been introduced to improve efficiency and scalability in long-horizon sequence modeling, while still learning representations from observation sequences. However, despite these advances, observations are primarily processed through latent representations or sequence models, and the importance of observation dimensions remains implicitly encoded within learned features rather than being explicitly modeled.

To address the lack of explicit modeling of observation importance, we propose adaptive importance metric learning (AIML), a modular component that learns trajectory-dependent importance weights over observation dimensions. This perspective enables the agent to adaptively focus on relevant observation dimensions, improving representation quality. Furthermore, by explicitly modeling observation importance, the proposed approach provides an interpretable characterization of how relevance is determined by context, offering insights into the agent's decision-making process.

Despite its advantages, the proposed approach presents several practical challenges. First, the computation of dimension-wise importance weights introduces additional computational overhead. Second, it introduces additional design choices, including the selection of sequence models and metric parameterization, which may affect optimization stability and require careful tuning across environments. Third, integrating the additional weighting mechanism into existing actor–critic frameworks can be challenging and may require architectural modifications. To address these challenges, we propose a lightweight module that integrates easily with standard actor–critic methods with minimal computational overhead.

Contributions The key contribution of this paper is the development and evaluation of AIML, a component that learns trajectory-dependent, dimension-wise importance weights and uses them to reweight observations. Our main contributions are as follows:

- We identify the limitations of fixed or implicitly learned observation weighting in reinforcement learning and motivate explicit modeling of observation relevance.
- We propose AIML, a modular component that explicitly learns importance weights over observation dimensions.
- We develop a simple and general algorithm that can be integrated into standard actor-critic methods with minimal architectural changes.

2 Problem setup

We consider observations $o_t \in \mathbb{R}^d$ at time step t, where d denotes the observation dimension. Each observation is represented as

$$o_t = (o_t^1, \ldots, o_t^d). \tag{1}$$

Bounded observations Each observation dimension lies within a range defined by the environment:

$$o_t^i \in [l_i, u_i], \quad i = 1, \ldots, d, \tag{2}$$

where l_i and u_i denote the lower and upper bounds. However, while the bounds l_i and u_i are structurally determined by the environment, the actual distribution of observed values within these bounds is generally unknown. Consequently, the lower and upper bounds are often specified or adjusted empirically.

Grouping of observations After specifying the bounds for each observation dimension, determining how to appropriately group observation dimensions remains challenging. Formally, let $\mathcal{G} = \{G_1, \ldots, G_K\}$ be a partition of $\{1, \ldots, d\}$, where each subset G_k represents a group of observation dimensions. In practice, the grouping $\mathcal{G}$ is typically determined based on domain knowledge, which may be suboptimal or task-dependent.

Heuristic weighting Even after grouping observation dimensions, determining the appropriate scale and relative importance of each dimension remains challenging. In practice, both the appropriate scale of weights for each observation dimension and their relative proportions across dimensions are often specified based on domain knowledge. As a result, such weighting schemes may be suboptimal or task-dependent.

$$\tilde{o} = w \odot o_t, \quad \sum_{i=1}^{d} w_i = 1, \quad w_i \geq 0, \tag{3}$$

where $w \in \mathbb{R}^d$ denotes the importance weights.

Reliance on empirical tuning As a result, identifying an effective combination of observation dimensions and their weighting schemes remains challenging. In practice, this process is typically addressed through empirical tuning, where parameters are iteratively adjusted based on observed performance:

$$(w, \mathcal{G}) \approx \arg\min_{w,\mathcal{G}} \mathcal{L}_{\text{empirical}}(w, \mathcal{G}), \tag{4}$$

where $\mathcal{L}_{\text{empirical}}$ denotes an implicit loss induced by empirical tuning, reflecting empirical performance such as observed returns, stability, or validation performance.

Partially observable environments These issues become more pronounced in partially observable settings, where observations are generated from a stochastic observation model,

$$o_t \sim p(o_t \mid s_t), \tag{5}$$

and the agent must infer latent states from incomplete observations.

A common approach is to maintain a belief over latent states,

$$b_t = p(s_t \mid o_{1:t}, a_{1:t-1}), \tag{6}$$

where a_t denotes the action taken at time step t. In such settings, inaccurate weighting of observation dimensions can degrade the quality of belief estimation.

3 Related work

Observation relevance in reinforcement learning. Several works have focused on extracting compact and informative representations from observation histories, aiming to mitigate partial observability and reduce redundancy in high-dimensional inputs.

The memory trace formulation [Eberhard et al., 2025] summarizes past observations using an exponential moving average, enabling agents to retain temporally aggregated information while avoiding the need to explicitly store full trajectories. Approaches for learning long-term representations [Yang et al., 2025] aim to preserve information relevant for long-term decision making while discarding irrelevant features, thereby improving generalization in visual reinforcement learning environments. By encouraging representations that focus on temporally consistent and decision-relevant signals, these methods help agents better capture the underlying structure of the environment beyond immediate observations. These methods primarily focus on learning latent representations that capture relevant information from observation histories, but may not explicitly expose the importance of individual observation dimensions in an interpretable manner.

A complementary line of work focuses on adaptive selection of observation components. An large language model (LLM) based active learning framework for partially observed and cost-aware settings [Astorga et al., 2024] sequentially selects which features or labels to acquire based on partially observed inputs, effectively identifying informative components of the observation. This approach highlights the importance of adaptive feature acquisition, where the selection process

is guided by contextual information inferred from observations. By leveraging LLMs to guide this process, the method enables flexible and context-dependent decisions on which information to acquire, demonstrating the potential of integrating language models into decision-making under partial observability. While this line of work focuses on selecting which information to acquire, it primarily operates through discrete acquisition decisions, rather than modeling continuous, dimension-wise importance within the policy or value function.

4 Preliminaries

In this section, we introduce AIML, which learns the importance of each observation dimension from historical context. We then describe how the proposed method can be integrated into standard actor-critic frameworks. To the best of our knowledge, this is among the first approaches to explicitly model dimension-wise importance over observations within actor-critic frameworks.

4.1 Adaptive Importance Metric Learning

Observation dimensions are normalized using environment-defined bounds, following standard practices in reinforcement learning.

$$\bar{o}_t^i = \frac{o_t^i}{\max(|l_i|, |u_i|)}, \tag{7}$$

where $\bar{o}_t^i$ denotes the normalized observation.

After normalization, to address the problem setup described earlier, we directly use the observations as input without introducing any manual grouping or predefined dimension-wise weighting.

To capture trajectory-dependent variations in observation importance, we model importance weights as a function of the observation history. Given an observation history $h_t = (\bar{o_1}, \ldots, \bar{o_t})$, we define dimension-wise importance weights as

$$w = (w_1, \ldots, w_d) = f(h_{N:t}) \in \mathbb{R}^d, \tag{8}$$

where f is a learnable function that maps the observation history $h_{N:t}$ to a set of importance weights. Each element w_i represents the importance of the corresponding observation dimension. Although the weights are computed from the observation history $h_{N:t}$, they are applied only to the current observation $\bar{o}_t$. We define the re-weighted observation as

$$\tilde{o} = w \odot \bar{o}_t, \tag{9}$$

where $\odot$ denotes element-wise multiplication. The transformed observation $\tilde{o}_t$ is then used by the policy and value functions.

4.2 AIML architecture

We now describe the implementation of the function $f(h_{N:t})$.

Sequence modeling To capture temporal dependencies, we use a sequence model over the observation history. In our experiments, we consider different sequence models, including single- and multi-head transformers and gated recurrent unit (GRU). The proposed framework is model-agnostic and can be applied to a wide range of sequence models (e.g., RNN, LSTM, Mamba). This flexibility arises from the fact that many sequence models produce outputs in a consistent temporal format, with sequences indexed from N to t. Consequently, different architectures can be integrated without modifying the overall framework. Given an observation history $h_t = (\bar{o_1}, \ldots, \bar{o_t})$, we process the sequence using a sequence model:

$$x_{N:t} = f_{\text{seq}}(\bar{o}_{N:t}) \in \mathbb{R}^{d_h}, \tag{10}$$

where f_{seq} denotes a sequence model, and $x_{N:t} = (x_N, \ldots, x_t)$ is the sequence of hidden representations with $x_k \in \mathbb{R}^{d_h}$ for $k = \{N, \ldots, t\}$.

Global query aggregation To summarize the hidden sequence $x_{N:t}$, we introduce a global query mechanism that identifies the most relevant timesteps in the observation history. We first generate a query from the last hidden representation:

$$q = f_q(x_t). \tag{11}$$

The query, keys, and values are then computed as

$$Q = W_Q q, \qquad K = W_K x_{N:t}, \qquad V = W_V x_{N:t}. \tag{12}$$

To improve numerical stability, we apply ℓ_2 normalization to the query and key representations before computing attention scores [Henry et al., 2020]:

$$\alpha = \mathrm{softmax}\left(Q' K'^{\top}\right), \tag{13}$$

where $Q' = \frac{Q}{\|Q\|_2}$ and $K' = \frac{K}{\|K\|_2}$. This normalization mitigates scale sensitivity in dot-product attention and improves the stability of the resulting softmax distribution.

Optionally, we apply a small smoothing term to prevent overly sharp attention:

$$\tilde{\alpha} = (1-\epsilon)\alpha + \frac{\epsilon}{\text{sequence length}}. \tag{14}$$

The global context vector is obtained by

$$c = \tilde{\alpha} V. \tag{15}$$

We refine the global context using residual connections, normalization, and a feedforward network:

$$r_1 = q + c, \tag{16}$$
$$n_1 = \mathrm{LayerNorm}_1(r_1), \tag{17}$$
$$n_2 = \mathrm{LayerNorm}_2(r_1 + \mathrm{FFN}(n_1)). \tag{18}$$

We then project the refined representation as

$$\tilde{c} = f_{\mathrm{proj}}(n_2). \tag{19}$$

Dimension-wise reweighting The global query module produces an aggregated history representation, denoted by $\tilde{c}$. We map this representation into a dimension-wise importance vector:

$$z = f_{\mathrm{head}}(\tilde{c}) \in \mathbb{R}^d. \tag{20}$$

To allow dimension-wise amplification or attenuation, we transform z as

$$\tilde{w} = \mathrm{softplus}(z). \tag{21}$$

which produces positive, unbounded importance weights.

To avoid excessively large scaling and improve numerical stability, we further constrain the weights using a bounded transformation:

$$w = \frac{\tilde{w}}{1 + \tilde{w}/\beta}, \tag{22}$$

where $\beta \geq 1$ controls the maximum scale of the weights (e.g., $\beta = 100$).

The resulting weights are then applied element-wise to the current observation:

$$\tilde{o} = w \odot \bar{o}_t \tag{23}$$

In this way, the importance of each observation dimension is determined from the observation history, while the reweighting itself is applied only to the current observation. This design avoids redundant transformations over past observations and ensures that the learned importance directly influences the decision at the current time step.

Output The reweighted observation $\tilde{o}$ is then passed through a projection and refinement block:

$$m = f_{\text{proj}}(\tilde{o}) \in \mathbb{R}^{d_h}, \tag{24}$$
$$r_1 = m + f_{\text{res}}(m), \tag{25}$$
$$n_1 = \text{LayerNorm}_1(r_1), \tag{26}$$
$$n_2 = \text{LayerNorm}_2(r_1 + \text{FFN}(n_1)). \tag{27}$$

Finally, the refined representation is projected into the output feature

$$aiml = f_{\text{out}}(n_2), \tag{28}$$

which is used as the final representation and consumed by the downstream policy and value networks. In addition to the final representation, the learned importance weights w can also be directly utilized as interpretable scaling factors over the observation dimensions. The overall architecture is illustrated in Appendix B.

5 Integration with Actor-Critic Methods

We integrate the proposed module into both the actor and critic networks. The actor and critic leverage a state representation s and the observation history h_t to construct trajectory-dependent representations.

At each time step t, both the actor and critic internally compute,

$$aiml = f(h_{N:t}). \tag{29}$$

The actor defines the policy as

$$\pi_\theta(a_t \mid [s, aiml]), \tag{30}$$

while the critic estimates the value as

$$V_\theta(s, aiml). \tag{31}$$

This design enables our proposed module to explicitly modulate the contribution of each observation dimension based on the observation history, while remaining independent of the state representation s. As a result, the policy and value functions can leverage both structured state estimation and adaptive observation reweighting, leading to improved robustness and more efficient learning under varying trajectory conditions.

6 Training Objective

We optimize the proposed model within a standard actor-critic framework.

Given trajectories $\tau = \{(o_t, a_t, r_t)\}_{t=1}^{T}$, the policy is trained to maximize the expected return.

We adopt the proximal policy optimization objective:

$$\mathcal{L}_\pi = \mathbb{E}_t \left[\min \left(r_t(\theta) A_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) A_t \right) \right], \tag{32}$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s, aiml_t)}{\pi_{\theta_{\text{old}}}(a_t|s, aiml_t)}$, A_t denotes the advantage estimate, and ϵ is a hyperparameter that controls the clipping range to limit the policy update.

The critic is trained to minimize the squared error:

$$\mathcal{L}_V = \mathbb{E}_t \left[\left(V_{\theta_V}(s, aiml_t) - R_t \right)^2 \right]. \tag{33}$$

The total loss is given by

$$\mathcal{L} = -\mathcal{L}_\pi - \beta \mathcal{H}(\pi) + \lambda \mathcal{L}_V, \tag{34}$$

where $\mathcal{H}(\pi)$ is the entropy bonus.

The overall objective is minimized with respect to the model parameters, enabling end-to-end learning.

7 Experiments and results

We evaluate AIML on a continuous control benchmark to assess its effectiveness in learning trajectory-dependent importance over observation dimensions. Our experiments focus on three aspects: overall performance, sample efficiency, and interpretability.

7.1 Experimental setup

Environments We conduct experiments on a standard control task, inverted double pendulum (MuJoCo) [Todorov et al., 2012]. This environment exhibits low-dimensional yet nonlinear dynamics, making it a suitable testbed for evaluating performance under varying levels of control complexity.

Baselines We compare the proposed approach against progressively enhanced variants. First, a multilayer perceptron (MLP) based actor–critic model using only the last observation h_t. Second, a sequence-based model with global query aggregation over the observation history $h_{N:t}$. Third, the full model, which further incorporates explicit dimension-wise importance weighting. To ensure a fair comparison, all models share the same network architecture except for the components replaced by the multilayer perceptron or sequence-based model. While the first two models rely on implicit representation learning, the proposed approach explicitly models observation importance. Details of the model implementation are provided in Appendix A.1.

Training To ensure a fair comparison, all methods share the same training configuration. We use the adam optimizer with a learning rate of 3×10^{-4}. The PPO clipping parameter is set to 0.2, the discount factor γ is 0.99, and generalized advantage estimation (GAE) [Schulman et al., 2017] λ is 0.95. We use a rollout length of 1000 steps and 8 parallel environments, resulting in 8000 transitions per update. The maximum reward per step is 1. Training is performed using mini-batches of size 256 sampled from the collected rollout data, and training is conducted for 500 episodes. Results are averaged over five random seeds. Both the actor and critic use networks with a hidden dimension of 128. We use a sequence length of 12, which captures the short-term dynamics of the inverted double pendulum task. Shorter sequences are insufficient to represent relevant observation history, while longer ones increase computational cost without clear benefits. The optimal sequence length is task-dependent and should be adjusted for different environments.

Evaluation metrics We evaluate all methods in terms of final performance, sample efficiency and return fluctuation.

Final performance is defined as the expected return:

$$R(\pi) = \mathbb{E}_{\tau \sim p_\pi} \left[\sum_{t=1}^{T} r_t \right]. \tag{35}$$

In practice, we approximate it by averaging episodic returns over the last 100 evaluation episodes out of 500 total training episodes, which provides a more reliable and fair estimate than using only a small number of evaluation episodes [Henderson et al., 2018].

Sample efficiency is measured by the average episodic return across all evaluation points, which approximates the area under the learning curve when evaluations are performed at fixed interaction intervals:

$$\frac{1}{K} \sum_{k=1}^{K} R_k, \tag{36}$$

where R_k is the episodic return at the k-th evaluation point and $K = 500$.

Return fluctuation is used to quantify training stability. Since episodic returns can increase over training, directly computing the standard deviation over all episodes may confound learning progress with instability. We therefore remove the local training trend using a moving average and compute the standard deviation of the residual returns:

$$\mathrm{Std}\left(\{R_k - \mathrm{MA}_{100}(R_k)\}_{k=100}^{K} \right), \tag{37}$$

where $\mathrm{MA}_{100}(R_k)$ denotes the moving average of the most recent 100 episodic returns and $K = 500$.

7.2 Results and discussion

We first compare non-sequential and sequential models within the AIML framework, while excluding the importance-weight computation. This comparison isolates the effect of temporal context by evaluating a non-sequential model against sequence-based variants.

Table 1: Performance comparison between non-sequential and sequential models over five random seeds. Higher final performance and sample efficiency are better, while lower return fluctuation indicates more stable learning.

Method	Final performance	Sample efficiency	Return fluctuation
MLP	953.59 ($\pm$ 9.21)	916.77 ($\pm$ 8.88)	31.63 ($\pm$ 2.63)
SHT	964.65 ($\pm$ 4.89)	931.93 ($\pm$ 5.78)	20.14 ($\pm$ 5.90)
MHT	954.68 ($\pm$ 10.01)	929.85 ($\pm$ 10.42)	26.82 ($\pm$ 13.88)
GRU	957.79 ($\pm$ 7.72)	931.43 ($\pm$ 11.88)	43.30 ($\pm$ 4.21)

From Table 1, sequence-based models that incorporate temporal context generally outperform the non-sequential baseline (MLP) in both final performance and sample efficiency. Among them, the single-head transformer (SHT) achieves the highest performance, while the multi-head transformer (MHT) and GRU show modest improvements. SHT also achieves the lowest return fluctuation, indicating more stable learning dynamics compared to the MLP baseline. MHT also reduces return fluctuation relative to MLP, although its variability across seeds is relatively large. In contrast, GRU exhibits higher return fluctuation than the MLP baseline, suggesting that incorporating temporal context through a recurrent hidden state may destabilize learning.

The effect of multi-head attention may generally depend on the dimensionality and diversity of the observation space. In low-dimensional observation spaces, decomposing temporal representations across multiple heads may provide limited benefit because each head has access to fewer distinct features for specialization. By contrast, higher-dimensional and more heterogeneous observations may allow different heads to specialize in different feature subspaces or dynamic patterns, potentially leading to stronger performance. In the present experiment, the observation space is relatively low-dimensional, consisting of only seven dimensions. This may limit the benefit of using multiple attention heads, since the implicit grouping induced by the heads may not provide sufficiently distinct or meaningful feature specialization. Moreover, such grouping may not always align with the underlying relevance of each observation dimension. As a result, the resulting representation may not fully capture trajectory-dependent importance, which can lead to limited performance improvements. To further examine this possibility, we provide an additional preliminary experiment on the higher-dimensional half cheetah task in Appendix C.2. The results provide preliminary evidence that multi-head attention may be more beneficial in richer and more heterogeneous observation spaces. For GRU, temporal information is compressed into a recurrent hidden state, which may make dimension-wise relevance less explicit and contribute to unstable learning dynamics.

Table 2: Performance comparison between sequence-based models and full models over five random seeds. Higher final performance and sample efficiency are better, while lower return fluctuation indicates more stable learning.

Method	Final performance	Sample efficiency	Return fluctuation
SHT	964.65 ($\pm$ 4.89)	931.93 ($\pm$ 5.78)	20.14 ($\pm$ 5.90)
SHT-AIML	955.33 ($\pm$ 15.99)	932.57 ($\pm$ 19.20)	20.96 ($\pm$ 8.21)
MHT	954.68 ($\pm$ 10.01)	929.85 ($\pm$ 10.42)	26.82 ($\pm$ 13.88)
MHT-AIML	951.79 ($\pm$ 20.42)	932.45 ($\pm$ 6.94)	20.14 ($\pm$ 13.45)
GRU	957.79 ($\pm$ 7.72)	931.43 ($\pm$ 11.88)	43.30 ($\pm$ 4.21)
GRU-AIML	967.03 ($\pm$ 3.75)	947.61 ($\pm$ 6.59)	13.26 ($\pm$ 4.46)

Building on these observations, we then examine the effect of incorporating the proposed method into different sequence-based architectures. The quantitative results are summarized in Table 2, and the corresponding episodic return curves are provided in Appendix A.3.

For transformer-based architectures, incorporating the proposed method does not improve final performance, but yields slightly higher sample efficiency and comparable or lower return fluctuation compared to the corresponding base models. In contrast, GRU-AIML improves across all metrics, achieving higher final performance and sample efficiency while substantially reducing return fluctuation.

These results suggest that the effectiveness of the proposed method depends on the underlying architecture. For transformer-based architectures, the proposed method may smooth the learning dynamics by reducing short-term return variation. At the same time, the decrease in final performance suggests a possible trade-off, as the additional reweighting operation may lead to more conservative learning dynamics and limit the improvement of final performance. One possible explanation is that transformer-based models already capture temporal dependencies to some extent, reducing the marginal benefit of the additional reweighting module. GRU-AIML shows clear improvement, suggesting that the method provides additional structure that is not sufficiently captured by the recurrent representation alone. Since GRU compresses temporal information into a single hidden state, some trajectory-dependent or dimension-wise information may be less explicitly preserved. The proposed method can be interpreted as helping to retain or emphasize such information, leading to improved performance.

Overall, these results indicate that the proposed method is more effective when the base architecture has limited capacity to represent trajectory-dependent structure, while its benefit is reduced when such structure is already partially captured. These observations also raise the possibility that the benefit of multi-head attention may depend on the dimensionality and heterogeneity of the observation space.

In addition to performance improvements, the proposed method offers the potential for interpretability by explicitly modeling dimension-wise importance. In actor-critic settings, the policy relies on a learned representation state s, which compresses observation history into a latent state. The learned weights w operate directly on the observation space, providing a potential lens to examine how the contribution of each observation dimension varies across trajectories, including aspects that may be underrepresented in s. We provide an illustrative example of the potential interpretability of the learned weighting patterns in Appendix D.

8 Summary and future work

Summary We proposed AIML, a simple and modular component that explicitly models trajectory-conditioned, dimension-wise importance over observations. AIML uses temporal context extracted from observation history to generate adaptive weights for each observation dimension and reweights the current observation accordingly. This design enables actor-critic models to incorporate trajectory-dependent information not only through latent sequence representations, but also through explicit observation-level weighting. Our results show that the effect of AIML varies across sequence-based architectures. The proposed module appears to achieve its strongest overall performance when combined with a base architecture that has limited capacity to represent trajectory-dependent structure. Beyond performance, the learned weights provide an explicit object for analysis, offering a potential path toward interpreting how the importance of each observation dimension changes across trajectories.

Limitations and future work Despite these promising results, several limitations remain. First, the empirical evaluation is limited to a single continuous control benchmark, which may restrict the generality of the findings. Future work should consider a broader range of environments, including tasks with higher-dimensional observations and more complex partial observability, and additional sequence modeling architectures. Second, although the proposed method mitigates computational overhead by leveraging existing sequence model structures, the additional cost of computing and applying dimension-wise weights remains. Future work should explore more efficient or lightweight designs to improve scalability. Finally, the current formulation applies reweighting only to observations at the current timestep. A promising direction is to apply a shared reweighting mechanism immediately after observation and use it as input to both the actor and critic. Furthermore, applying the same dimension-wise weighting mechanism to belief state representations is an important direction for future work. In addition, establishing a more systematic framework for evaluating and utilizing the potential interpretability of the learned weighting patterns is an important direction for future research.

Acknowledgments

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References

Juan Astorga et al. Active learning with llms for partially observed and cost-aware scenarios. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. Decision transformer: Reinforcement learning via sequence modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

Onno Eberhard, Michael Muehlebach, and Claire Vernade. Partially observable reinforcement learning with memory traces. In *International Conference on Machine Learning (ICML)*, 2025.

Farama Foundation. Inverted double pendulum. `https://gymnasium.farama.org/environments/mujoco/inverted_double_pendulum/`, 2024. Accessed: 2026-04-27.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.

Matthew Hausknecht and Peter Stone. Deep recurrent q-learning for partially observable mdps. *arXiv preprint arXiv:1507.06527*, 2015.

Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.

Alex Henry, Prudhvi Raj Dachapally, Shubham Pawar, and Yuxuan Chen. Query-key normalization for transformers. *arXiv preprint arXiv:2010.04245*, 2020.

Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

Maximilian Igl, Luisa Zintgraf, Tuan Anh Le, Frank Wood, and Shimon Whiteson. Deep variational reinforcement learning for POMDPs. In *International Conference on Machine Learning (ICML)*, 2018.

Steven Kapturowski, Georg Ostrovski, John Quan, Rémi Munos, and Will Dabney. Recurrent experience replay in distributed reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2019.

Qi Lv, Xiang Deng, Gongwei Chen, Michael Yu Wang, and Liqiang Nie. Decision mamba: A multi-grained state space model with self-evolution regularization for offline rl. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL `https://proceedings.neurips.cc/paper_files/paper/2024/hash/288b63aa98084366c4536ba0574a0f22-Abstract-Conference.html`.

Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, and Georg Ostrovski. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.

John Schulman, Filip Wolski, Prafulla Dhariwal, et al. Proximal policy optimization algorithms. In *arXiv preprint arXiv:1707.06347*, 2017.

Emanuel Todorov, Tom Erez, and Yuval Tassa. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033. IEEE, 2012.

Rui Yang, Jie Wang, Qijie Peng, Ruibo Guo, Guoping Wu, and Bin Li. Learning robust representations with long-term information for generalization in visual reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2025.

A Training details

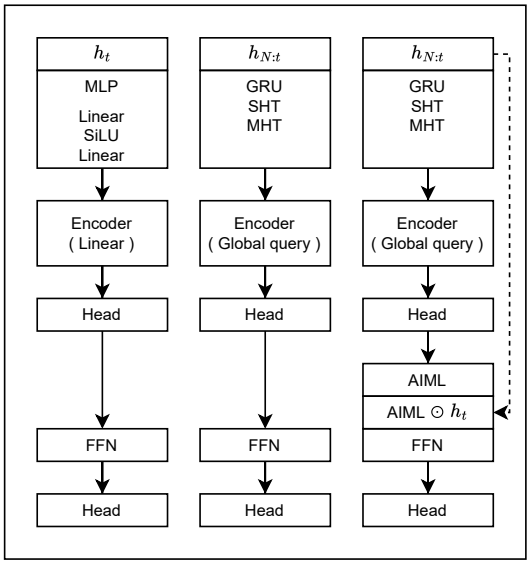


Figure 1: Comparison of implementation variants.

A.1 Implementation

Figure 1 presents the implementation variants considered in this work, including configurations with and without the proposed AIML module. The figure illustrates how AIML is integrated into different sequence-based architectures and allows for consistent comparisons across model configurations. For a fair comparison, the MLP-based model uses only the last time step while maintaining an input structure that is as similar as possible to those of the sequence-based models, ensuring a comparable experimental setting.

A.2 Hyperparameters

Table 3: Training hyperparameters

Hyperparameter	Value
Learning rate	3×10^{-4}
PPO clip	0.2
Discount γ	0.99
GAE λ	0.95
Train batch size	256 (32 per environment $\times$ 8)
Episode	500
Rollout length	1000
Parallel environment	8
Random seed	10, 20, 30, 40, 50 (each offset by environment index 0-7)

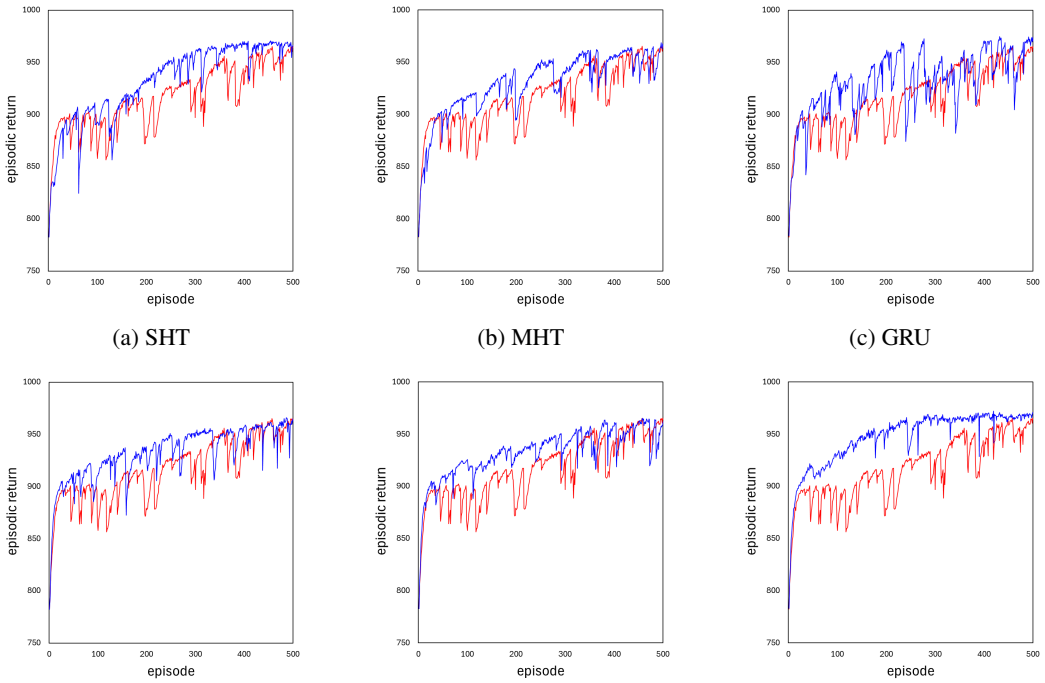
Table 4: AIML hyperparameters

Hyperparameter	Value
Global query smoothing ϵ	0.02
Maximum β	100
Sequence length	12
Hidden dim	128 (scaled to 4$\times$ input dim when input dim exceeds 128)

Table 3 summarizes the general training hyperparameters used across all experiments, including the optimizer configuration, rollout length, and batch composition. The batch size is constructed from multiple parallel environments, ensuring efficient data collection during training.

Table 4 presents the specific hyperparameters of the proposed AIML module, including the scaling range of the learned weights and the sequence modeling configuration.

A.3 Additional results and learning curves



(a) SHT (b) MHT (c) GRU

(d) SHT + AIML (e) MHT + AIML (f) GRU + AIML

Figure 2: Comparison of episodic return, where the MLP-based model is shown in red and the sequence-based models are shown in blue.

Figure 2 compares the episodic return curves of sequence-based models with and without AIML in the inverted double pendulum task. The top row shows the base sequence models, and the bottom row shows their AIML-augmented counterparts. The red curves denote the MLP-based baseline, while the blue curves denote the corresponding sequence-based models. Since the maximum per-step reward is 1.0 and each episode consists of 1000 steps, the maximum episodic return is 1000. All methods are trained for 500 episodes and evaluated over five random seeds.

A.4 Open access and computational resources

Our implementation is provided via an anonymous repository at `https://anonymous.4open.science/r/Implementation_2026-0DD7/`, which is included as supplementary material for review purposes.

All experiments were conducted using an NVIDIA RTX 4060 GPU with 8GB of memory. Each episode takes approximately 20 seconds, and training consists of 500 episodes per run. Training a single model for one random seed takes approximately 166 minutes under standard conditions.

B AIML framework

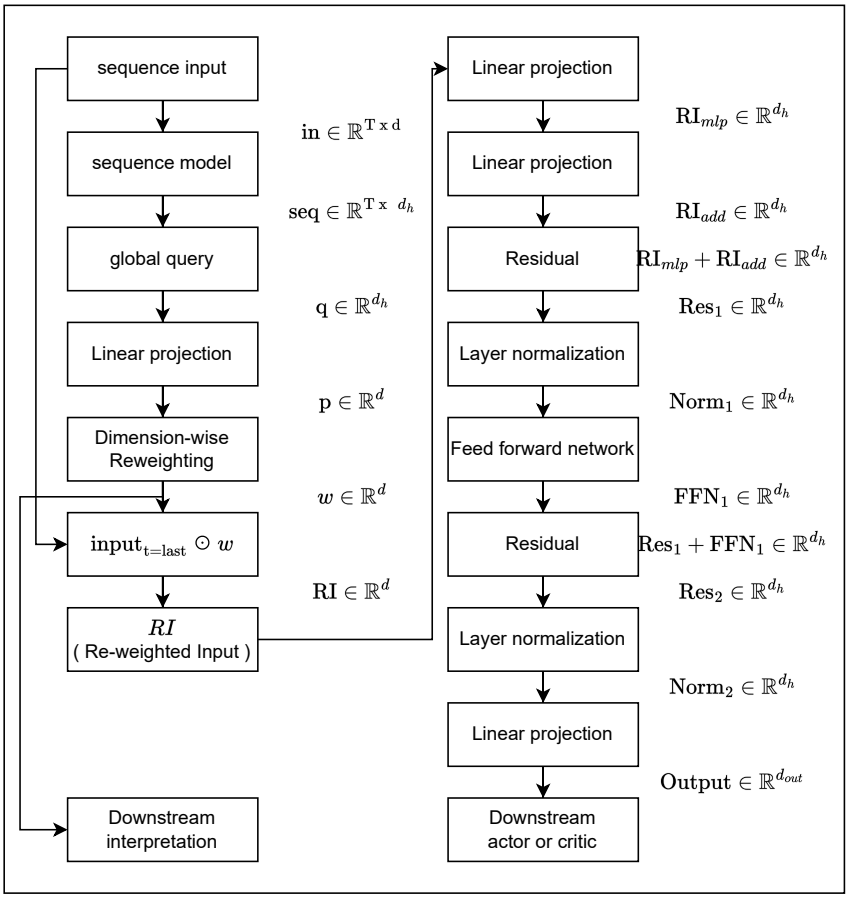


Figure 3: Overview of the AIML framework.

This section provides an overview of the AIML framework.

As illustrated in Figure 3, the model first encodes the observation history using a sequence-based module. The resulting temporal context is then used to generate trajectory-conditioned, dimension-wise importance weights. These weights are applied to the current observation, allowing the policy and value networks to emphasize observation dimensions that are more relevant under the current trajectory.

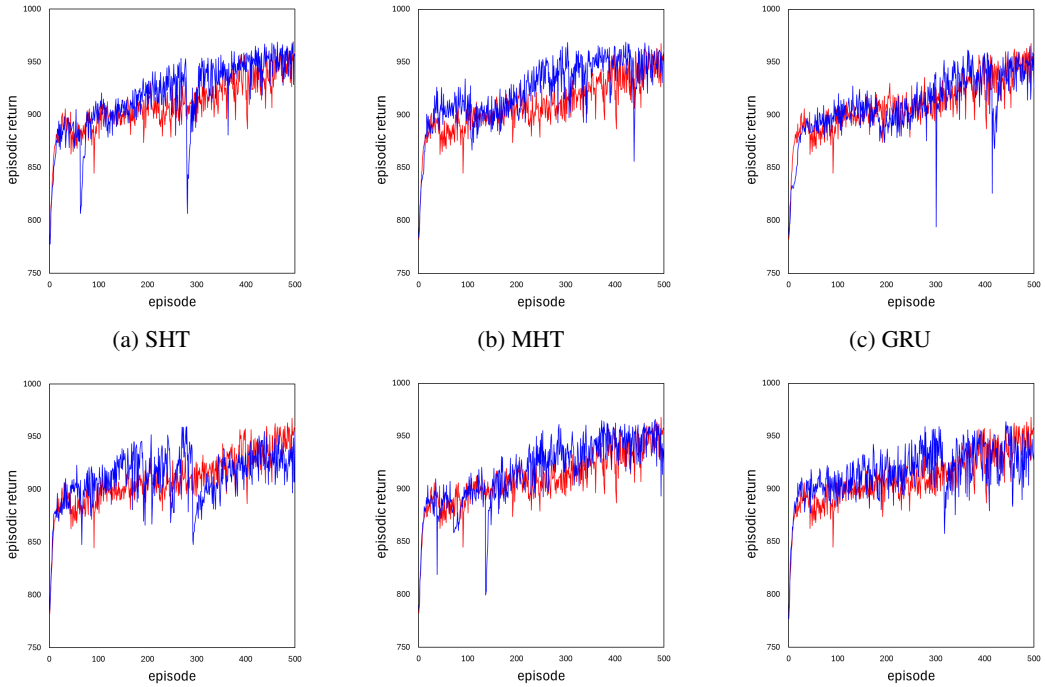
In addition, the learned weights w can be used outside the module for interpretability purposes. For example, in environments with unstructured observations, w can serve as an indicator for identifying which observation dimensions are emphasized when the agent selects actions that lead to higher expected returns.

C Comparison

C.1 Initial design

In the initial design of the AIML framework, normalization was not applied to the query-key projections in the transformer and global query modules. For the dimension-wise reweighting, the weights were constrained to the range $[1, \beta]$ to preserve the baseline observation, and a rounding operation was applied to w for additional stabilization.

However, as shown in Figure 4, in the inverted double pendulum task, incorporating temporal context did not consistently lead to the expected performance improvements over non-sequential settings. Furthermore, models incorporating AIML exhibited unstable training dynamics. To address this issue, we introduce ℓ_2 normalization in the query-key projections to improve the stability of attention computation. In addition, we revise the reweighting formulation to allow both amplification and attenuation by expanding the range to $[0, \beta]$. We also remove the rounding operation, as we found that continuous scaling provides more flexible and context-dependent modulation than discrete adjustments.



(a) SHT (b) MHT (c) GRU

(d) SHT + AIML (e) MHT + AIML (f) GRU + AIML

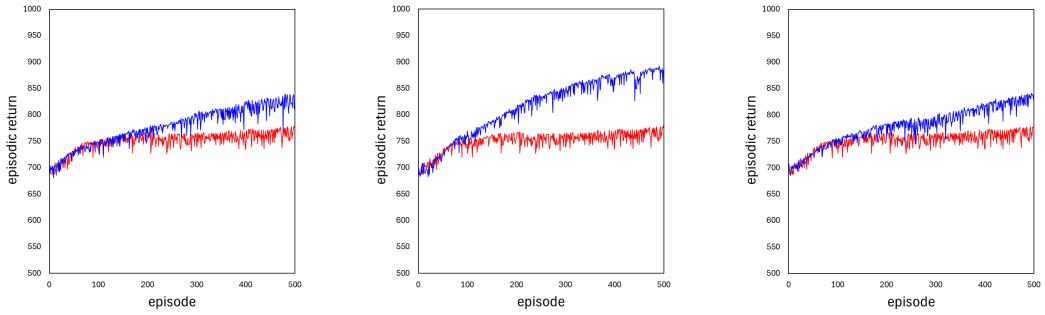
Figure 4: Comparison of episodic return under the initial design, where the MLP-based model is shown in red and the sequence-based models are shown in blue.

C.2 Higher dimensional observations

To further examine the relationship between multi-head attention and AIML in higher-dimensional observation spaces, we conduct an additional preliminary experiment in the half cheetah MuJoCo environment [Todorov et al., 2012]. This environment is a continuous control benchmark in which a two-dimensional cheetah-like robot learns to move forward by coordinating multiple joints. Compared with the inverted double pendulum task, it has a higher-dimensional observation space. In the standard MuJoCo setting used in this experiment, the observation vector consists of 17 dimensions, including joint positions and velocities. This makes the task suitable for examining whether multi-head attention and AIML behave differently when the observation space contains richer and more heterogeneous information. Although the preliminary experiment uses only one random seed, 10, all evaluation metrics, including final performance, sample efficiency, and return fluctuation, are computed using the same criteria as those used in the main experiments. The training and AIML hyperparameters are also kept identical to those used in the main experiments.

Table 5: Preliminary results on the half cheetah task with higher-dimensional observations.

Method	Final performance	Sample efficiency	Return fluctuation
MLP	763.90	750.19	9.90
SHT-AIML	819.68	777.26	9.84
MHT-AIML	876.46	815.00	11.58
GRU-AIML	823.63	778.71	10.15



(a) SHT-AIML (b) MHT-AIML (c) GRU-AIML

Figure 5: Episodic return curves on the half cheetah task. The MLP baseline is shown in red, and each AIML-based sequence model is shown in blue.

Table 5 presents preliminary results on the higher-dimensional half cheetah task. The results show that all AIML-based sequence models outperform the non-sequential MLP baseline in terms of final performance and sample efficiency. Among them, MHT-AIML achieves the highest final performance and sample efficiency, suggesting that multi-head attention may be more beneficial when the observation space is higher-dimensional and contains more heterogeneous information.

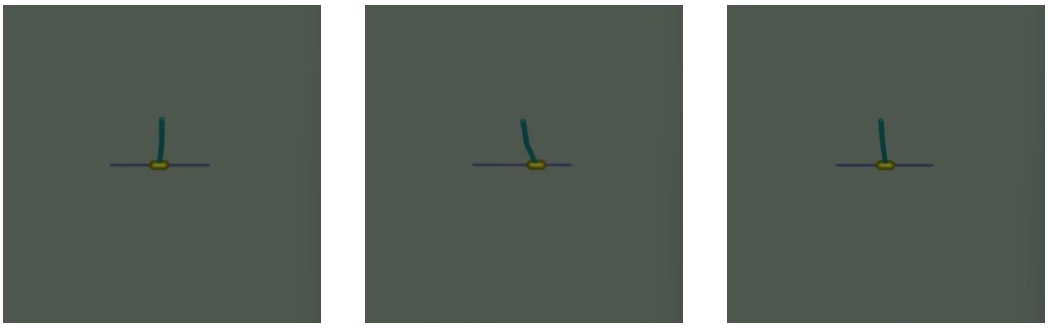
This result is consistent with the intuition that multiple attention heads can specialize in different feature subspaces or dynamic patterns when the observation vector provides sufficiently diverse information. In contrast to lower-dimensional settings, where multi-head decomposition may provide limited benefit, the richer observation structure in half cheetah may allow MHT-AIML to better exploit trajectory-dependent and dimension-wise information. Moreover, the fact that all AIML-based sequence models improve over the non-sequential baseline suggests that the AIML framework can also operate effectively in higher-dimensional observation spaces. However, since this experiment is preliminary, further evaluation with multiple random seeds is needed to draw stronger conclusions.

D Interpretability

In the inverted double pendulum task, the observation vector is defined as shown in Table 6, following the gymnasium implementation [Farama Foundation, 2024].

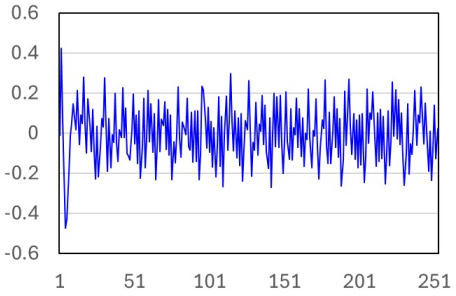
Table 6: Observation components in the inverted double pendulum task.

Dimension	Description
`cart_position`	x_c (cart position)
`pole1_sin`	$\sin(\theta_1)$ (sine of pole 1 angle)
`pole2_sin`	$\sin(\theta_2)$ (sine of pole 2 angle)
`pole1_cos`	$\cos(\theta_1)$ (cosine of pole 1 angle)
`pole2_cos`	$\cos(\theta_2)$ (cosine of pole 2 angle)
`cart_velocity`	v_c (cart velocity)
`pole1_angular`	ω_1 (angular velocity of pole 1)
`pole2_angular`	ω_2 (angular velocity of pole 2)

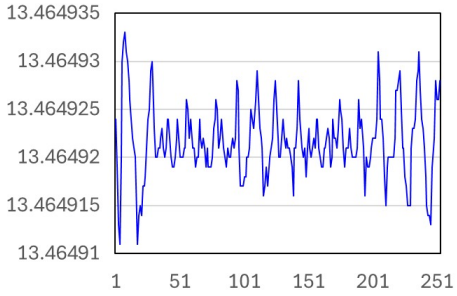


(a) State a (b) State b (c) State c

Figure 6: Example states of the inverted double pendulum.



(a) Input angular velocity (`pole1_angular`), and x-axis is rollout step, y-axis is value.



(b) Learned weight w for angular velocity, and x-axis is rollout step, y-axis is value.

Figure 7: Temporal evolution of angular velocity and corresponding learned weights across trajectories.

We examine the learned dimension-wise weights w in relation to the observation components defined in Table 6.

We observe that the weights corresponding to `cart_position`, `pole1_sin`, `pole2_sin`, `pole1_cos`, `pole2_cos`, and `cart_velocity` are close to zero. Since the belief state is defined to model the next observation, this suggests that it may already capture sufficient information about these components.

In contrast, the weights for `pole1_angular` and `pole2_angular` are significantly amplified. This suggests that angular velocity components receive substantially higher importance, potentially reflect-

ing their critical role in the control dynamics of the task. It is worth noting that the magnitude of this amplification is not absolute and may vary depending on the normalization of the observation inputs.

For example, as shown in Figure 6 (a), the normalized angular velocity is -0.472288 (rad/s), where the negative sign indicates clockwise rotation, with a corresponding weight of 13.464920.

As the system transitions to Figure 6 (b), the angular velocity decreases to -0.012374 (rad/s), while the corresponding weight is slightly increased to 13.464933, suggesting that the policy adjusts the weighting during action selection in response to changes in the state.

As the system further transitions to Figure 6 (c), the angular velocity becomes 0.090609 (rad/s), with a corresponding weight of 13.464921, indicating that the system approaches a more stable state.

A broader view of these dynamics is provided in Figures 7.

Overall, these results suggest that the learned weights selectively emphasize observation dimensions that may be underrepresented in the belief state, providing a potential lens for interpretability by revealing how dimension-wise importance evolves across trajectories and contributes to state stabilization.

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